AI-CONTROLLED HIGH-PERFORMANCE UNCREWED AIRCRAFT WITH CONFIDENCE-GATED ADAPTIVE FLIGHT MANAGEMENT

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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to [PROVISIONAL APPLICATION NUMBER/DATE], if applicable. If no priority application is used, this paragraph should be deleted or revised before filing.

FIELD OF THE INVENTION

[0002] The present disclosure relates to uncrewed aircraft systems and, more particularly, to high-performance uncrewed aircraft having onboard artificial-intelligence-assisted flight management, multimodal sensor fusion, adaptive health-aware control, communications resilience, cryptographically authenticated human supervision, mission-state reconciliation, and auditable evidence logging.

BACKGROUND

[0003] Uncrewed aircraft are increasingly capable of operating with limited direct control. Conventional architectures often separate sensing, navigation, mission planning, aircraft health monitoring, communications, and flight control into software layers with limited shared state. Such separation can create inconsistent assumptions when sensor confidence changes, components degrade, communications are interrupted, or a remote operator modifies a mission during flight.

[0004] Existing autonomous flight systems may also rely on a single mission plan generated before launch, or on planners that optimize a route without continuously coupling route selection to aircraft health, sensor confidence, communications quality, and verified operating constraints. A technical need exists for an integrated onboard architecture that converts uncertain measurements into a controlled flight state while constraining AI-generated behavior through deterministic safety logic and preserving verifiable mission history.

SUMMARY OF THE INVENTION

[0005] In one aspect, the invention provides an uncrewed aircraft comprising an airframe, heterogeneous sensors, an onboard compute subsystem, a state-estimation engine, an environment model, a mission-contract interpreter, an artificial-intelligence planner, a deterministic safety-envelope engine, a health-state estimator, a flight-control interface, a communications manager, an evidence logger, and a reconciliation engine.

[0006] The state-estimation engine receives heterogeneous measurements and produces a fused state together with confidence information and provenance information. The environment model

represents environmental features separately from aircraft state and can associate uncertainty with those features. Confidence and provenance may be propagated into planning so that candidate actions are not treated as unqualified commands.

[0007] The artificial-intelligence planner generates one or more candidate trajectories or navigation behaviors from a mission contract and the current state. A forward evaluator 213 can predict aircraft response over a planning horizon. The deterministic safety-envelope engine evaluates a candidate against protected operating constraints, environment uncertainty, sensor validity, communications state, and current health state, and can accept, modify, reject, or replace the candidate before it reaches the flight controller.

[0008] Predicted aircraft-health information can update planning constraints before trajectory selection. A health-state estimator can use telemetry, vibration, electrical, thermal, actuator, and control-residual information to estimate degradation. The planner can then select a trajectory having sufficient margin for the predicted condition while maintaining recovery options.

[0009] Human-originated instructions can be represented as cryptographically authenticated, capability-scoped mission contracts. The aircraft verifies the contract, including issuer, validity interval, action classes, operational constraints, and revocation state, before exposing the contract instructions to the planning subsystem. A communications manager can detect degraded link conditions and limit executable behavior to previously authenticated action classes under a configured degraded policy.

[0010] During degraded communications, a reconciliation engine maintains a local mission-state record and, after link recovery, compares the local record with a remote record using sequence identifiers, timestamps, authority information, provenance information, and conflict status. An unresolved conflict is prevented from becoming executable until resolved according to the applicable authority policy or reviewed by an authorized supervisor.

[0011] An evidence logger stores selected state transitions, model identifiers, confidence information, health state, command provenance, communications events, and safety-envelope decisions. The log may be append-only or tamper-evident through hash chaining or another integrity mechanism.

[0012] These features may be implemented independently or in combinations and may improve autonomy reliability, controlled adaptation, communications resilience, auditability, and traceability in high-performance uncrewed aircraft.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 illustrates an example uncrewed aircraft platform and principal onboard subsystems.

[0014] FIG. 2 illustrates an onboard compute and autonomy architecture including state estimation, planning, health management, supervision, communications, and evidence functions.

[0015] FIG. 3 illustrates heterogeneous sensor fusion, confidence assessment, provenance tracking, and an environment model with uncertainty.

[0016] FIG. 4 illustrates AI candidate generation, forward evaluation, deterministic safety-envelope gating, and delivery of an accepted or modified action to flight control.

[0017] FIG. 5 illustrates health prediction and health-aware reconfiguration in which predicted degradation updates planning constraints before trajectory selection.

[0018] FIG. 6 illustrates authenticated human supervision and resilient communications with degraded-link policy handling.

[0019] FIG. 7 illustrates mission-state reconciliation using local and remote state records, authority, sequence, timestamp, provenance, and conflict status.

[00110] FIG. 8 illustrates onboard compute, model lifecycle controls, and evidence logging with integrity protection.

DETAILED DESCRIPTION

Architecture and airframe

[0020] FIG. 1 depicts an uncrewed aircraft 100 having an airframe 102, propulsion system 104, flight-control surfaces 106, power system 108, communications subsystem 110, heterogeneous sensor suite 112, onboard compute subsystem 114, and storage subsystem 116. The airframe may be fixed-wing and may have a high-performance configuration, although the inventive features are not limited to a particular aerodynamic layout. The propulsion system may include one or more engines or electric propulsion units. The aircraft may include non-kinetic payload interfaces for sensing, navigation, communications relay, environmental monitoring, research, or other lawful functions.

[0021] The onboard compute subsystem 114 may include one or more general-purpose processors, graphics processors, field-programmable gate arrays, dedicated safety controllers, or combinations thereof. Functions described as separate modules may be combined, replicated, partitioned, or distributed while preserving the claimed control relationships.

Onboard autonomy stack

[0022] FIG. 2 shows sensor ingest 200, feature extraction 202, state estimation 204, environment model 206, mission-contract interpreter 208, AI planner 210, safety-envelope engine 212, health-state estimator 214, flight-control interface 216, communications manager 218, evidence logger 220, and reconciliation engine 222. A mission contract 230 can provide authenticated objectives, permitted action classes, temporal validity, geographic or operational constraints, and revocation state. A human supervisory station 240 can originate or approve mission contracts and receive evidence records.

Sensor fusion, confidence, provenance, and environment uncertainty

[0023] FIG. 3 illustrates fusion of inertial, air-data, navigation, optical, infrared, radar, lidar, acoustic, or other measurements. A measurement may be associated with source identity, acquisition time, calibration version, quality score, and integrity status. The state-estimation engine 204 can produce a fused state vector including position, velocity, attitude, angular rates, energy state, and associated confidence or uncertainty information. Environment features may be represented separately and can carry uncertainty estimates.

[0024] Confidence information may be propagated from source measurements through state estimation and planning. Provenance may identify sensor sources, timestamps, calibration or estimator versions, model versions, and relevant state snapshots. A confidence policy may require corroboration, alternate sensing, human confirmation, or a fallback mode when confidence is insufficient.

Mission contracts

[0025] The mission-contract interpreter 208 can receive a machine-readable mission contract 230 authenticated by a trusted authority. The contract may contain a mission identifier, issuer identifier, validity interval, permitted action classes, operational constraints, resource limits, geographic constraints, and revocation state. The interpreter can verify a digital signature and authority chain before exposing instructions to the planner.

AI planning, forward evaluation, and deterministic safety gating

[0026] FIG. 4 illustrates an AI planner 210 generating multiple candidate trajectories or navigation behaviors. Candidate generation may use model-predictive control, learned policies, reinforcement learning, search, heuristic planning, optimization, or hybrid techniques. A forward

evaluator may simulate or approximate predicted aircraft response over a short horizon using the candidate, current state, health state, and environment model.

[0027] The safety-envelope engine 212 independently evaluates the candidate against deterministic constraints associated with structural limits, propulsion limits, control authority, energy reserves, sensor validity, environment uncertainty, communications state, authorized operational limits, and current health state. The safety-envelope engine may accept a candidate, modify it to satisfy constraints, reject it, select another candidate, or cause a verified fallback behavior. Candidate outputs from the AI planner are not sent directly to actuators. An independent safety controller may override an AI output when a protected constraint would be violated.

Health-aware adaptation

[0028] FIG. 5 shows the health-state estimator 214 receiving telemetry such as motor or engine data, vibration, electrical measurements, thermal measurements, actuator feedback, and control residuals. A statistical or machine-learning model may estimate a fault class, degradation probability, remaining useful operation interval, or other health indicator. The health information updates one or more planning constraints before trajectory selection or control.

[0029] For example, when predicted degradation reduces expected maneuver authority or increases energy consumption, the planner may select a trajectory that maintains larger margins, reduces control demand, preserves recovery options, or changes resource allocation. Mode-transition logic can include persistence or hysteresis conditions to avoid rapid oscillation among operating modes.

Communications resilience and authenticated supervision

[0030] FIG. 6 shows the communications manager 218 monitoring link quality, latency, packet loss, authentication state, and channel availability. The manager can select among available links or relay paths. When authenticated command connectivity degrades, a configured degraded policy may permit only previously authorized action classes while newly received or partially received commands remain non-executable until integrity and authority are re-established.

[0031] The human supervisory station 240 can issue, modify, suspend, or revoke mission contracts. The aircraft may verify authority before accepting a mission update. A revocation or other higher-authority state can supersede an otherwise valid but stale local action according to the reconciliation policy.

Mission-state reconciliation

[0032] FIG. 7 shows the reconciliation engine 222 maintaining a local mission-state record and comparing it with a remote mission-state record after communication recovery. State records can include monotonic sequence numbers, timestamps, authority level, provenance, validity state, and conflict state. The reconciliation engine applies deterministic rules to merge compatible records, reject superseded or stale commands, elevate authoritative revocations, and preserve unresolved conflicts for review rather than making them executable.

Model lifecycle and evidence logging

[0033] FIG. 8 illustrates a model package 236 having a model identifier, compatibility metadata, validation status, signing information, and authorized deployment scope. A model update can be admitted only when a trusted policy verifies required metadata and deployment authorization. A rollback model may be retained.

[0034] The evidence logger 220 can record selected state transitions, model identifiers, confidence data, health estimates, command provenance, communications events, safety-envelope decisions, and model activation state. An integrity record 238 may be append-only or hash-chained, and a rollback model 239 may be retained to support controlled reversion. The record can permit later reconstruction of the sequence of candidate actions and corresponding acceptance, modification, or rejection decisions without requiring retention of every raw sensor sample.

Distributed implementation, shadow evaluation, training, and alternatives

[0035] The functions can be distributed among multiple onboard processors, a companion aircraft, a ground station, or a remote compute service. The aircraft can retain local enforcement of protected safety constraints even if remote services become unavailable. A shadow evaluator may assess alternative models without directly controlling the aircraft and may record comparison results for later validation.

[0036] Training data can include historical flight data, synthetic simulation data, controlled test data, and fault-injection scenarios. Validation may assess state-estimation error, trajectory tracking, constraint-violation rates, recovery performance, latency, and confidence calibration. The disclosed functions may also be implemented with different machine-learning architectures, deterministic estimators, sensor combinations, or communications protocols.

Industrial applicability

[0037] The disclosed architecture can be used in high-performance uncrewed aircraft for training, flight research, reconnaissance, search and rescue, communications relay, environmental monitoring, logistics, or other lawful missions requiring reliable autonomous flight and controlled human supervision. The disclosure does not require weapon targeting, weapon release, or lethal engagement functionality.

REFERENCE NUMERALS

100 aircraft; 102 airframe; 104 propulsion system; 106 flight-control surfaces; 108 power system; 110 communications subsystem; 112 heterogeneous sensor suite; 114 onboard compute subsystem; 116 storage subsystem; 200 sensor ingest; 202 feature extraction; 204 state-estimation engine; 206 environment model; 208 mission-contract interpreter; 210 AI planner; 212 safety-envelope engine; 213 forward evaluator; 214 health-state estimator; 216 flight-control interface; 218 communications manager; 220 evidence logger; 222 reconciliation engine; 230 mission contract; 240 human supervisory station.

CLAIMS

1. An uncrewed aircraft system comprising: an airframe; a plurality of heterogeneous sensors; an onboard compute subsystem; a state-estimation engine configured to generate a fused aircraft state from outputs of the heterogeneous sensors and to associate the fused aircraft state with confidence information and provenance information; an environment model configured to represent at least one environment feature with associated uncertainty information; an artificial-intelligence planner configured to generate a plurality of candidate flight trajectories or navigation behaviors based at least in part on the fused aircraft state, the environment model, and a machine-readable mission contract; a forward evaluator configured to predict a response of the aircraft to at least one candidate; a deterministic safety-envelope engine configured to evaluate the candidate against protected operating constraints including at least one constraint dependent on the confidence information, the uncertainty information, or an aircraft health state; and a flight-control interface configured to expose to an aircraft controller an action only after the candidate has been accepted, modified, or replaced by the deterministic safety-envelope engine.

2. The uncrewed aircraft system of claim 1, wherein the provenance information identifies one or more source sensors, acquisition times, calibration information, estimator versions, model versions, or state-estimation snapshots associated with a component of the fused aircraft state.

3. The uncrewed aircraft system of claim 1, further comprising a health-state estimator configured to infer a predicted degradation condition of at least one aircraft subsystem from telemetry, and wherein the predicted degradation condition updates one or more planning constraints before the artificial-intelligence planner selects a trajectory for execution.

4. The uncrewed aircraft system of claim 3, wherein the artificial-intelligence planner selects among at least a first candidate trajectory and a second candidate trajectory according to the predicted degradation condition and a remaining operating margin.

5. The uncrewed aircraft system of claim 1, wherein the deterministic safety-envelope engine is configured to accept a candidate, modify the candidate to satisfy a protected constraint, reject the candidate, or select an alternative candidate.

6. The uncrewed aircraft system of claim 1, wherein the deterministic safety-envelope engine includes or communicates with an independently computed safety controller configured to prevent an action that violates a protected aircraft limit.

7. The uncrewed aircraft system of claim 1, wherein the mission contract includes a mission identifier, issuer identifier, validity interval, authorized action classes, one or more operating

constraints, and revocation information, and wherein the mission-contract interpreter verifies a digital signature or authority chain before the artificial-intelligence planner receives an instruction from the mission contract.

8. The uncrewed aircraft system of claim 1, further comprising a communications manager configured to determine a degraded communications state and, while in the degraded communications state, restrict executable actions to a set of previously authenticated action classes.

9. The uncrewed aircraft system of claim 8, further comprising a reconciliation engine configured to maintain a local mission-state record during the degraded communications state and, after communications recovery, compare the local mission-state record with a remote mission-state record using at least sequence information, time information, authority information, and provenance information.

10. The uncrewed aircraft system of claim 9, wherein the reconciliation engine prevents an unresolved conflict between the local mission-state record and the remote mission-state record from becoming executable.

11. The uncrewed aircraft system of claim 9, wherein the reconciliation engine gives precedence to an authorized revocation over a stale command according to a defined authority policy.

12. The uncrewed aircraft system of claim 1, further comprising an evidence logger configured to record at least a model identifier, a state snapshot, confidence information, aircraft health information, command provenance, and a safety-envelope decision.

13. The uncrewed aircraft system of claim 12, wherein the evidence logger produces an append-only or tamper-evident record using hash chaining or another integrity mechanism.

14. The uncrewed aircraft system of claim 12, wherein the evidence logger records candidate trajectory identifiers and corresponding acceptance, modification, rejection, or fallback outcomes.

15. The uncrewed aircraft system of claim 1, wherein a model package used by the artificial-intelligence planner includes a model identifier, compatibility metadata, validation status, signing information, and an authorized deployment scope, and wherein a trusted policy verifies the model package before activation.

16. The uncrewed aircraft system of claim 15, wherein the onboard compute subsystem retains a rollback model and activates the rollback model when the activated model fails a configured validation or integrity condition.

17. The uncrewed aircraft system of claim 1, wherein at least one candidate trajectory is evaluated in a shadow mode in which a predicted response is generated without directly commanding the aircraft.

18. The uncrewed aircraft system of claim 1, wherein a fallback flight mode is selected when confidence information falls below a threshold, when protected constraints cannot be satisfied, or when authenticated command connectivity is lost.

19. The uncrewed aircraft system of claim 1, wherein the onboard compute subsystem implements the state-estimation engine, the environment model, the artificial-intelligence planner, the safety-envelope engine, and the health-state estimator using two or more computational resources selected from a central processor, graphics processor, field-programmable gate array, and independent safety controller.

20. A computer-implemented method of controlling an uncrewed aircraft, the method comprising: receiving heterogeneous sensor measurements; generating a fused aircraft state having confidence information and provenance information; generating an environment model having uncertainty information; receiving or validating a machine-readable mission contract; generating, using an artificial-intelligence planner, a plurality of candidate trajectories or navigation behaviors; predicting aircraft response for at least one candidate; generating a predicted aircraft health condition; updating one or more planning or safety constraints based on the predicted aircraft health condition before trajectory selection; deterministically evaluating a candidate against the updated constraints; and commanding a flight controller only with an action that satisfies the deterministic safety evaluation.

21. The method of claim 20, further comprising detecting degraded communications, restricting executable behavior under a degraded-link policy, preserving a local mission-state record, and after communication recovery reconciling the local mission-state record with a remote mission-state record using authority, sequence, time, provenance, and conflict information.

22. The method of claim 21, further comprising preventing an unresolved mission-state conflict from becoming executable and giving precedence to an authorized revocation over a stale command.

23. The method of claim 20, further comprising recording model identity, state, confidence, health, command provenance, communications state, and safety-envelope decisions in a tamper-evident event record.

24. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors of an uncrewed aircraft, cause the processors to perform the method of claim 20.

ABSTRACT OF THE DISCLOSURE

An uncrewed aircraft includes heterogeneous sensors, a confidence-aware state-estimation engine, an environment model with uncertainty, a mission-contract interpreter, an artificial-intelligence planner, a forward evaluator, a deterministic safety-envelope engine, a health-state estimator, a flight-control interface, a resilient communications manager, a reconciliation engine, and an evidence logger. Sensor measurements are fused into an aircraft state having confidence and provenance information while environment features carry uncertainty information. The planner generates candidate trajectories or navigation behaviors, and the forward evaluator predicts aircraft response. A deterministic safety envelope evaluates the candidate against protected operating constraints and predicted aircraft health before an action reaches flight control. Health information can update planning constraints before trajectory selection. Human instructions can be authenticated through capability-scoped mission contracts. During communications degradation, executable behavior can be restricted to previously authenticated actions while local and remote mission-state differences are preserved for later reconciliation. Evidence can include model identity, state, confidence, health, command provenance, communications state, and safety decisions.